

Angel Rodriguez

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EDUCATION

University of Central Florida

Orlando, FL

Bachelor of Science in Computer Science

Expected Graduation: Spring 2029

Relevant Coursework: C Programming, Python Automation, Object Oriented Programming (Java)

Extracurricular Activities: ColorStack, SHPE, and KnightHacks

EXPERIENCE

Software Engineering Intern

Jan 2026. – April. 2026

Minoria Tech

Orlando, FL

- Developed a suite of **RESTful APIs** using **FastAPI** and **SQL**, replacing third-party CRM dependencies and improving system control and scalability
- Designed and implemented authentication, database architecture for enterprise clients including Lockheed Martin, Dell, and Red Hat
- Collaborated in a Git-based team environment, contributing to code reviews and API design documentation to ensure maintainability and efficient onboarding

NASA L'Space NPWEE Member

Jan. 2026 – March. 2026

L'SPACE

Remote

- Selected for NASA's competitive L'Space program, contributing to a mission proposal focused on low-temperature Sabatier reactors for Mars ISRU fuel production
- Collaborated on the design of a Nickel-Manganese catalyst system to convert CO₂ and H₂ into methane (CH₄), targeting a reduction in operating temperature from ~400°C to ~250°C for improved energy efficiency
- Contributed to technical analysis and documentation evaluating CO₂ conversion efficiency, catalyst performance, and cost reduction to support long-duration Mars mission sustainability

Robotics Team Lead Programmer

Sep. 2023 – June. 2025

Botball

New Jersey

- Led programming team of **10** for an autonomous robot using C and KIPR Wombat Controller, enabling full task automation in competition scenarios
- Integrated sensors, camera vision, servos, and motor control systems to achieve **75% task success rate** in autonomous navigation and object manipulation challenges
- Developed and tested modular C code for obstacle navigation and object handling, improving execution consistency and reliability across competition runs

PROJECTS

DSL – AI Humanitarian Crisis Intelligence Platform | *Python, FastAPI, React, SQL*

- Placed **3rd out of 24** in Georgia Tech's Databricks Challenge for AI-driven humanitarian funding analysis
- Built a FastAPI + Pandas backend to analyze humanitarian datasets and compute funding gaps and crisis metrics
- Developed a React frontend with interactive dashboards and an AI assistant RAG Bot for real-time data exploration

Python Information Aggregator App | *Python*

- Built a Python CLI application integrating multiple external APIs to aggregate real-time weather, tech, news, and sports data
- Designed a modular architecture with JSON parsing and automated email delivery for daily information summaries
- Implemented error handling and rate-limit management to ensure reliable API consumption and consistent data retrieval

Full-Stack Weather Web Application | *JavaScript, HTML, CSS*

- Built responsive web app integrating OpenWeatherMap API
- Implemented asynchronous API calls and dynamic DOM rendering
- Optimized user experience with conditional rendering and responsive UI design for mobile and desktop views

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, HTML/CSS

Frameworks/Libraries: Flask, FastAPI, React, Node.js, pandas, OpenCV, PyTorch, Tailwind CSS

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Eclipse

AI Tools: Claude/Codex/Cursor, RAG